- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 5) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 60 samples (6 lots) of MOSFETs (FDFMA3N109). The 50 Gy TID exposure reduced the voltage threshold of the MOSFETs by around 15 mV. The NIEL exposure did not affect the functionalities of the irradiated devices.
- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 6a and 6b) and **NIEL** tests were done in 20 samples (2 lots) of Instrumentation amplifiers (INA333) configured with gain ~500. Linearity and Gain stability were stable over the TID tests and verified after the NIEL exposure. The performance of the devices was not affected by the radiation.
- **TID** and **NIEL** tests were done in 10 Direktronik SFP+ samples (batch 20122113 of model 33-4248) with no impact on the performance of the samples
- **SEE** tests were performed in 3 Direktronik SFP+ samples that integrated: **TID: 167 Gy, NIEL: 2.61x10E12 (1 MeV n)/cm2,** and **High Energy Hadrons fluence: 7.78x10E11**
  - **TID** effects and **SEL** occurrences were not detected during the current monitoring of the devices over the irradiation period. The currents were stable and the devices functioned well over 12 power cycles (Fig. 7a).
  - **SEU** rates were studied over 8 runs (Fig. 7b and 7c), where average rates between **1x10xE-8 errors/(h cm2)** and **8x10xE-8 errors/(h cm2)** were measured over all the runs for the three irradiated modules
    - Accounting for the average rate of **5.67x10E-8 errors/(h cm2)** (worst performing module), a **total of 13440 errors per SFP+** can be expected over 10 years of HL-LHC
    - **9.15 bit errors/min** are expected in all TileCal, which will be mitigated by the **GBTx GBT-FEC** protocol capable of handling 16 consecutive errors every 25 ns.
  - Two rare occurrences of what seems to be **SEFI-like** behaviour happened during the runs and were recovered via power cycling
    - The rate of the SEFI-like occurrence is **0.65 SEFI/day** for a total of **2389 SEFI** over the whole HL-LHC period
    - These events will be further studied, but wont impact the data taking since can be easily managed by power cycles, resets and the redundancy layers.
- **TID** to **50 Gy @ 3.42 Gy/h** (Fig. 8a and 8b) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 10 samples (1 lot) of FLASH memories (IS25LP256). The exposure did not impact the behaviour and current consumption of any of the functionalities of any of the irradiated devices. The NIEL exposure did not affect the functionalities of the irradiated devices.
- **TID** to **50 Gy @ 3.24 Gy/h** (Fig. 9a and 9b) and **NIEL to 5x10E12 (1 MeV n)/cm2** tests were done in 40 samples (4 lots) of 40 MHz and 100 MHz oscillators. The current consumption and the frequencies of all the devices were stable over the whole TID exposure. The NIEL exposure did not affect the functionalities of the irradiated devices.

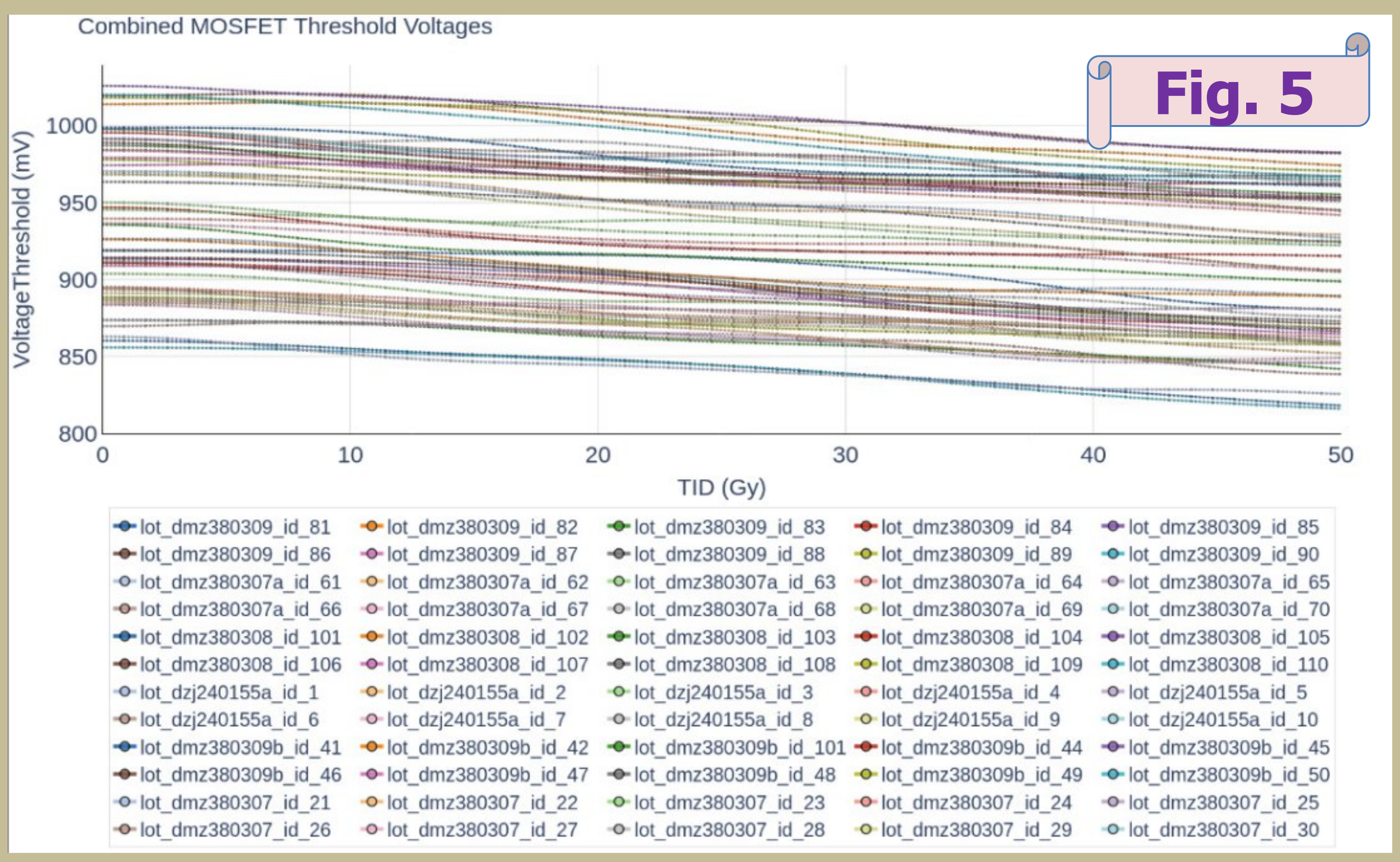

Fig. 5

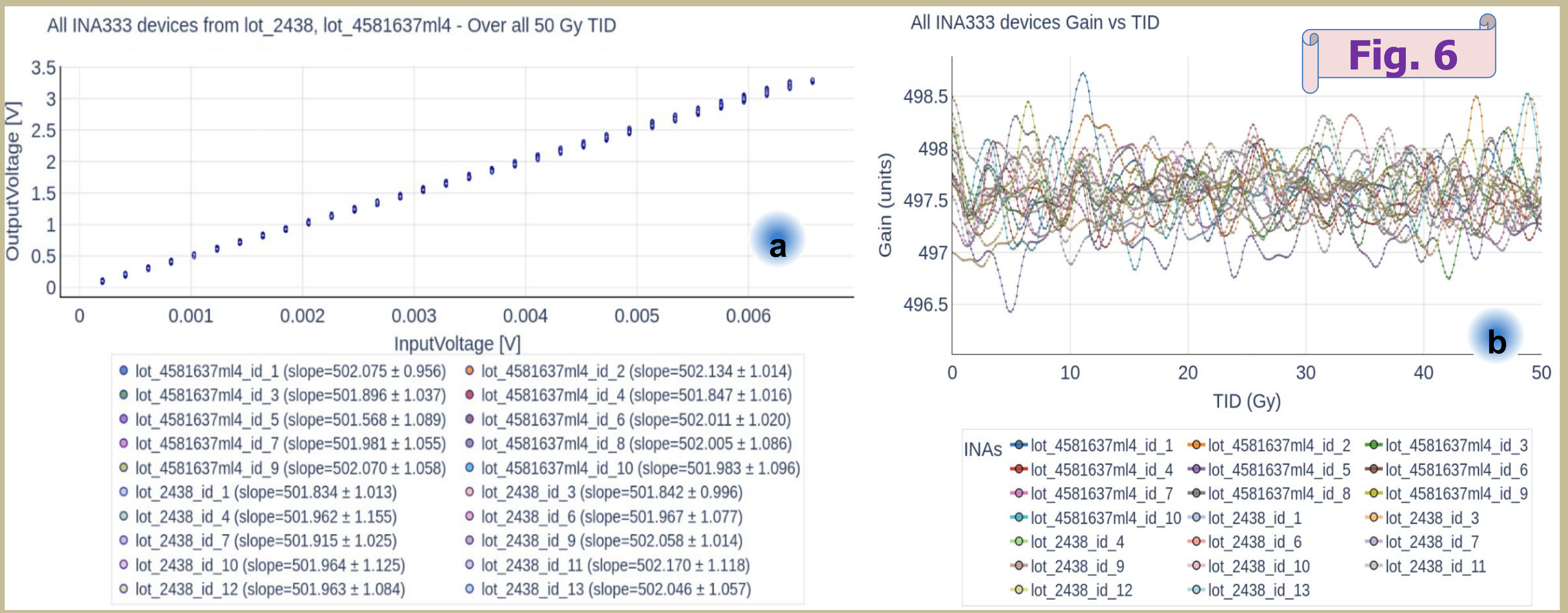

Fig. 6

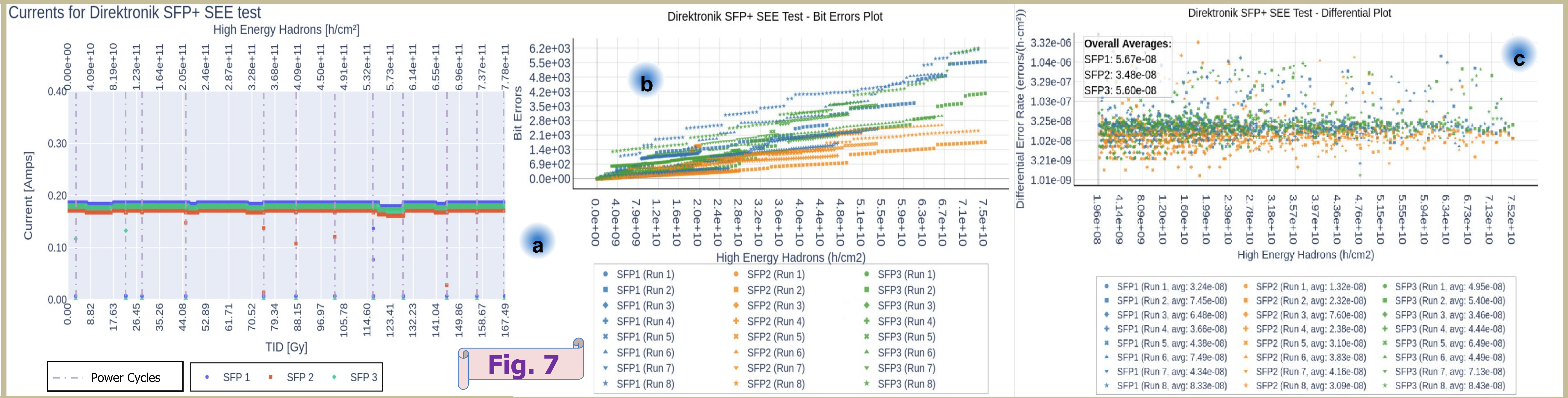

Fig. 7

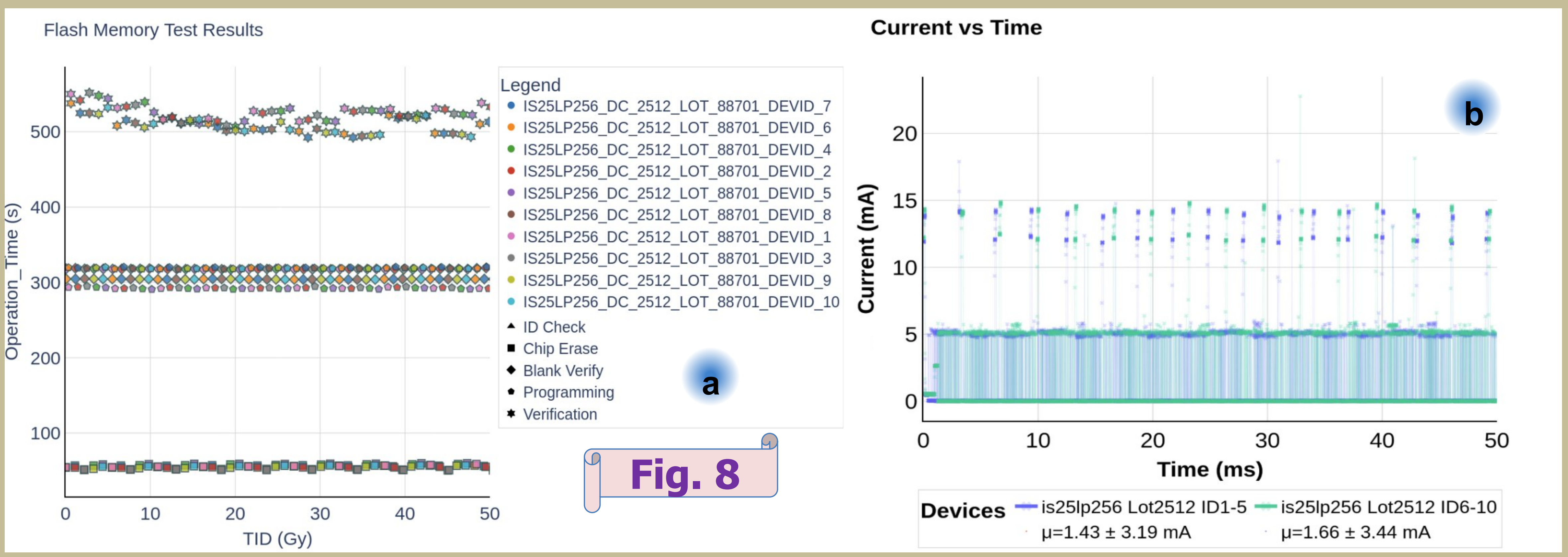

Fig. 8

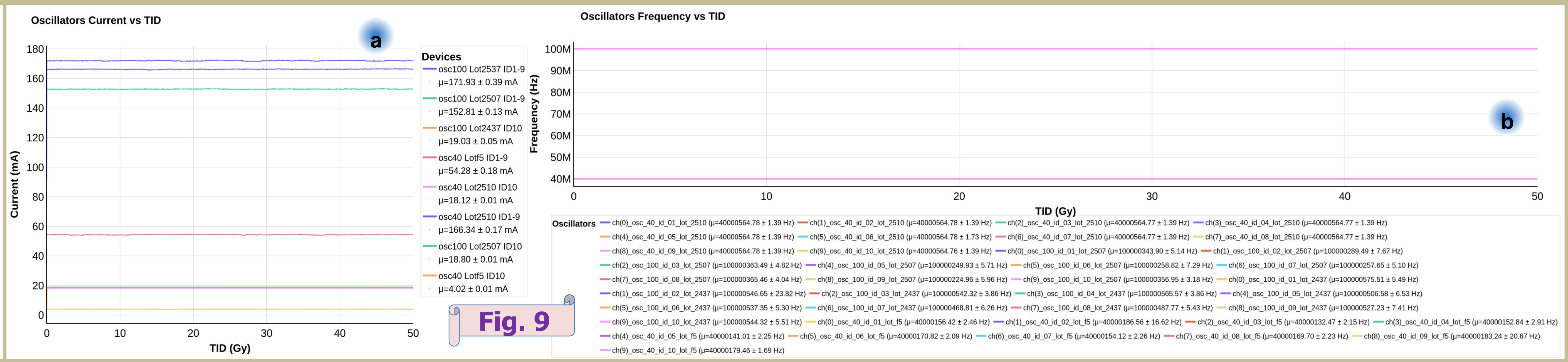

Fig. 9